

Ankit

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M.Sc. student in Computational Mathematics & Data Science with hands-on experience in machine learning, deep learning, API development, and data analysis. Passionate about building production-ready ML systems for real-world problems.

Education

National Institute of Technology, Rourkela	Aug 2025 – Present
M.Sc. in Computational Mathematics and Data Science (Certificate)	CGPA: 7.95
Sh. Rameshwar Lal Sharma Teacher Training College, Pilani	2021 – 2025
B.Sc. – B.Ed (Degree)	CGPA: 7.23

Projects

- Heatwave Prediction in Rajasthan** 🔗
- Python, FastAPI, Scikit-learn, Pandas, NumPy, Docker*
- Engineered a production-ready ML system to predict heatwave conditions and assess outdoor work safety, achieving 92.5% classification accuracy.
 - Developed 5 RESTful API endpoints using FastAPI with Pydantic validation for real-time predictions, weather fetching, and risk assessment.
 - Built feature engineering pipeline incorporating temperature–humidity interactions (heat index) and auto-fetched OpenWeather API data.
 - Deployed RandomForestClassifier generating risk-stratified outputs (Low/Medium/High) with binary heatwave predictions and work advisories.
 - Containerized with Docker for reproducible deployment; integrated frontend UI for end-to-end prediction workflow.
- Sleep Breathing Analysis** 🔗
- Python, TensorFlow, Keras, NumPy, SciPy, Matplotlib*
- Built an end-to-end pipeline to classify sleep breathing events (Hypopnea, Obstructive Apnea) from PSG signals across 5 participants.
 - Applied 4th-order Butterworth bandpass filter (0.17–0.4 Hz), segmented into 30-second windows with 50% overlap, and trained a 1D CNN evaluated via Leave-One-Participant-Out Cross-Validation.
 - Generated per-participant PDF diagnostic reports with annotated PSG signal overlays and confusion matrices.

Academic Lab Work

- Numerical Analysis Laboratory** 🔗
- Aug 2025 – Nov 2025
- Python, NumPy, SciPy*
- Implemented algorithms for nonlinear equations, linear systems, and ODE models
 - Performed convergence and error analysis for numerical stability
 - Benchmarked efficiency of iterative computational methods
- Statistics Laboratory** 🔗
- Jan 2026 – Present
- C++*
- Built statistical computation programs from scratch in C++, demonstrating low-level programming proficiency without relying on standard libraries
 - Implemented simulation models for probability distributions
 - Computed descriptive and inferential statistics efficiently

Technical Skills

Programming: Python | **APIs:** FastAPI
Data Analysis: Pandas, NumPy, SciPy, Matplotlib, Seaborn, EDA, Statistical Modelling
Machine Learning: Scikit-learn, Regression, Decision Trees, Ensemble Methods, Model Evaluation
Deep Learning: TensorFlow, Keras, 1D CNN
Time Series Analysis: ARIMA, SARIMA, Prophet, Forecasting, Trend & Seasonality Decomposition
Database: MySQL **Tools:** Git, GitHub, Docker

Certifications

Python Essentials – Cisco 2025